WEATHER TREND FORECASTING

Data analysis • anomaly detection • predictive modelling • spatial comparison

PROJECT REPORT
Dataset: GlobalWeatherRepository.csv
Coverage: 16 May 2024 – 10 March 2026
Forecasting case study: Vatican City

PM Accelerator mission alignment
PM Accelerator's official website describes its PMA KIDS initiative as providing free product-management education to teenagers from underserved families. Its stated mission is to break down financial barriers and achieve educational fairness, with an ambition to establish 200 schools worldwide over 20 years. This report reflects that education-access mission through a structured, evidence-led data-science case study.

This is an independent weather-data project report, not an official PM Accelerator product. Mission summary: pmaccelerator.io.

1. Executive Summary

This project explores global weather observations through data preparation, exploratory data analysis (EDA), trend analysis, anomaly detection, forecasting-model comparison, feature analysis, and geographical comparisons. The dataset contains **128,638 records and 41 columns**, covering 16 May 2024 to 10 March 2026. The forecasting case study focuses on Vatican City.

The reported Ridge Regression–XGBoost ensemble achieved MAE 2.476, RMSE 3.225, and R^2 0.152 on a common 127-observation test period. The previously reported tuned Random Forest achieved MAE 3.2221, RMSE 3.9371, and R^2 −0.2639. The ensemble improves on that baseline, but its R^2 remains modest; forecasts should not be presented as highly reliable.

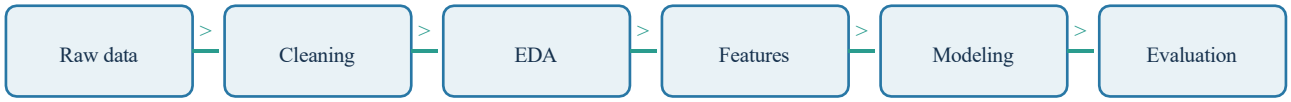
Project facts

Item	Reported detail
Dataset size	128,638 rows × 41 columns
Date coverage	16 May 2024 – 10 March 2026
Forecast location	Vatican City
Common test period	127 observations
Leading reported model	Ridge Regression–XGBoost ensemble

2. Data and Methodology

The source file is GlobalWeatherRepository(1).csv. The workflow uses Python/Jupyter, Pandas and NumPy for data handling, Matplotlib/Seaborn for visualization, Scikit-learn and XGBoost for machine learning, and Statsmodels for time-series analysis.

2.1 End-to-end workflow



2.2 Data cleaning

The cleaning stage should validate data types, parse date/time fields, inspect missing values and duplicates, check numeric ranges, and review inconsistent observations. A cleaning stage is part of the known notebook workflow. Exact per-column missing-value counts, duplicate counts, outlier rules, and before/after row totals were not available for this report and are not invented.

2.3 Feature preparation

Date-derived and meteorological features are prepared for analysis and forecasting. Time-series evaluation should preserve chronological order. The reported redesigned comparison uses a common test window, making candidate-model metrics more comparable.

3. Exploratory Data Analysis (EDA)

The EDA scope includes distributions and relationships among temperature and other weather variables, temporal patterns, correlation analysis, and comparisons across locations. These views help identify data quality issues, variability, possible seasonality, and geographic differences.

EDA component	Purpose
Variable distributions	Inspect central tendency, spread, skew, and extreme values for temperature, humidity, wind, pressure, precipitation, and related measures.
Correlation analysis	Explore associations among weather variables; correlation alone does not establish causation.
Temporal analysis	Examine changes over time and possible seasonal or short-term patterns.
Geographic comparisons	Compare countries and continents while considering differences in coverage and sample sizes.

Visualization note

The original notebook's rendered plots were not accessible during report generation. This PDF does not fabricate dataset-specific charts. For a fully evidenced appendix, export the notebook's actual temperature time series, distributions, correlation heatmap, anomaly plots, regional comparisons, and feature-importance visualizations.

4. Trend Analysis and Anomaly Detection

Trend analysis examines changes in weather variables over time. Anomaly detection flags observations that depart from typical patterns. Such observations may represent genuine extreme weather, local conditions, sensor/reporting errors, or data-quality problems. They should be investigated in context rather than automatically removed.

Interpretation cautions

The roughly 22-month dataset is too short by itself to establish long-term climate change. Global aggregates can conceal local variation. Anomaly flags are investigative signals, not proof of erroneous data or a climate event.

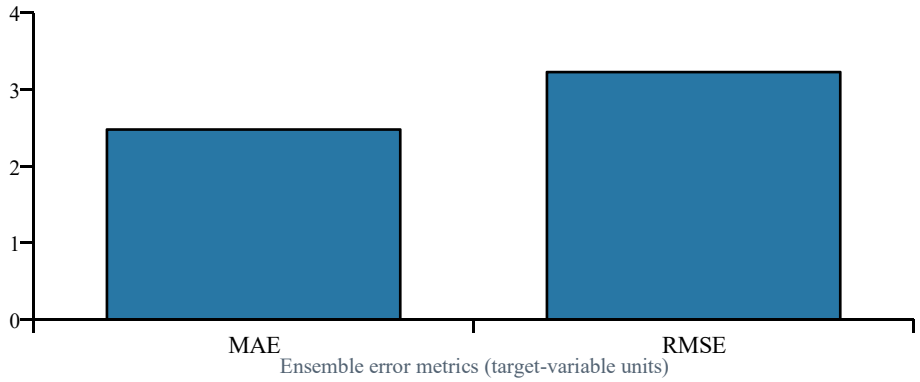
5. Forecasting Models

The project compares Linear Regression, Random Forest, XGBoost, ARIMA, Exponential Smoothing, and a Ridge Regression–XGBoost ensemble. These cover linear, nonlinear tree-based, and classical time-series approaches.

Model	Role
Linear Regression	Interpretable linear baseline; may miss nonlinear structure.
Random Forest	Tree ensemble for nonlinear relationships and interactions.
XGBoost	Gradient-boosted trees; requires tuning and leakage-aware validation.
ARIMA	Classical autoregressive/integrated/moving-average time-series model.
Exponential Smoothing	Time-series baseline for level and optional trend/seasonality.
Ridge–XGBoost ensemble	Combines a regularized linear model with boosted trees.

6. Model Evaluation

Model	MAE	RMSE	R^2
Ridge–XGBoost ensemble	2.476	3.225	0.152
Tuned Random Forest	3.2221	3.9371	–0.2639



Ensemble error metrics (target-variable units)

Relative to the reported tuned Random Forest, ensemble MAE is approximately 23.2% lower and RMSE approximately 18.1% lower. R^2 improves from negative to 0.152, but remains low.

Metric interpretation

MAE is the average absolute prediction error; RMSE penalizes larger errors more strongly. R^2 = 0.152 means the model explains about 15.2% of target variance under the standard R^2 definition. It is not a 15.2% accuracy score. Error units depend on the target variable's scale.

A robust evaluation should also include a naive or seasonal-naive baseline, rolling-origin validation, residual diagnostics, and performance by month/location. Full scores for every candidate model were not available and are not fabricated.

7. Advanced Analyses

Feature importance

Feature-importance analysis examines which inputs contribute to model predictions. Importance is model-dependent and does not prove causal influence. Correlated predictors can distort rankings; permutation importance or SHAP-based explanations on held-out data can add context.

Spatial and regional analysis

Country- and continent-level comparisons describe geographic variation in observed weather. Interpretation should account for unequal record counts, differing observation periods, station coverage, elevation, and local climate. Report sample sizes and dispersion alongside averages.

Climate and environmental context

Climate summaries can describe patterns within the available observation window, but the period is insufficient for formal long-term climate-normal or climate-change attribution. Environmental-impact claims require additional contextual evidence.

8. Key Insights and Recommendations

Finding	Implication / action
Ensemble errors are lower than tuned RF errors.	Keep as experimental candidate; validate independently.
R^2 is 0.152.	Communicate uncertainty; compare against simple baselines.
Weather varies by place and time.	Evaluate by location and season, not only pooled metrics.
Dataset period is short for climate inference.	Acquire longer historical records before long-horizon claims.
Anomaly flags require context.	Review observations before excluding them.

9. Limitations and Future Work

Limitations

• The original notebook and CSV were not available for direct inspection during PDF generation; detailed cleaning counts and original plot images could not be verified.
• The dataset covers a limited period for climate inference.
• The ensemble's R^2 is modest.
• A 127-observation test set may not represent all weather regimes.
• Feature importance and geographic comparisons are descriptive unless supported by stronger causal/sampling analysis.

Recommended next steps

1. Re-run the notebook end-to-end from a clean kernel and save outputs.
2. Add a cleaning audit table with missingness, duplicates, invalid values, and retained row counts.
3. Publish full common-test metrics for every model, including a naive baseline.
4. Add rolling-origin validation and fold-level variability.
5. Plot actual vs predicted values and residuals; inspect error by month and weather regime.
6. Document target variable, units, forecast horizon, features, and ensemble weights.
7. Export actual EDA, anomaly, feature-importance, and spatial plots into the report.

Conclusion

The project establishes an end-to-end workflow for weather-data exploration and forecasting. The Ridge–XGBoost ensemble is the strongest specifically documented comparison, but the modest R^2 indicates substantial unexplained variation. It is best described as a promising experimental baseline, not a high-confidence operational forecasting system.

10. Tools and References

Project assets: GlobalWeatherRepository.csv; implementation_final.ipynb.
Technology: Python, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, XGBoost, Statsmodels, Jupyter.
PM Accelerator mission source: Official PM Accelerator website. The homepage describes PMA KIDS' free product-management education initiative for underserved teenagers and its goal of educational fairness.

Report provenance: Dataset dimensions, coverage, forecast location, model metrics, and analysis scope are based on project details previously supplied in this conversation. Because the source notebook and CSV were not retrievable for this report run, this document distinguishes reported results from analysis details that still need direct notebook evidence.